RDD2022: A Multi-National Image Dataset for Automatic Road Damage Detection

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Abstract: Road damage identification is essential for pavement management, vehicle maintenance, and preventing fatal accidents. Machine learning algorithms trained on single-country datasets frequently fail to generalize across diverse global regions due to differences in road materials, vehicle mounts, camera resolutions, and weather. This study introduces RDD2022, the largest multi-national pavement distress benchmark compiled from crowdsensing initiatives across six nations: Japan, India, the Czech Republic, Norway, the United States, and China. The dataset contains 47,420 high-resolution road images with 55,007 meticulously annotated bounding box instances spanning four core distress classes: longitudinal cracking (D00), transverse cracking (D10), alligator cracking (D20), and potholes (D40), alongside extended regional distress categories. We provide baseline detection evaluations across one-stage and two-stage object detectors, establishing public benchmark criteria for international road condition assessment.

Dataset Characteristics & Distribution Summary

• **Total Images:** 47,420 multi-national photographs (38,385 training, 9,035 evaluation).
• **Geographic Scope:** 6 nations (India: 9,215 images; Japan: 10,506; Czech: 3,365; Norway: 8,161; US: 5,006; China: 11,167).
• **Distress Taxonomy:** D00 (Wheel mark longitudinal cracks), D10 (Equal-interval transverse cracks), D20 (Alligator fatigue mesh), D40 (Potholes), D43 (Crosswalk blur), D44 (Lane line blur), D01/D11 (Severe rutting), D50 (Manhole utility defect).

Baseline Benchmark Detection Performance (mAP@50)

Baseline Detector	Japan (mAP)	India (mAP)	Czech (mAP)	Norway (mAP)	US (mAP)	China (mAP)	Global mAP@50
YOLOv5s	56.2%	51.4%	53.8%	49.6%	52.1%	54.7%	52.8%
YOLOv7	61.4%	56.8%	58.2%	54.1%	57.3%	59.8%	57.9%
YOLOv8x	65.8%	61.2%	62.4%	58.9%	62.0%	64.1%	62.4%
Faster R-CNN	62.1%	55.4%	57.0%	52.8%	56.4%	58.1%	56.9%